

Machine Learning Model Outcomes

Executive summary

Overview

The New York City Taxi & Limousine Commission (TLC) has engaged the Automatidata data team to develop a machine learning model aimed at predicting whether a taxi passenger will leave a generous tip.

Problem

Due to ethical concerns, the team decided to shift the modeling objective from predicting non-tippers to predicting “generous” tippers, defined as those who leave a tip of 20% or more. This approach balances the interests of both taxi drivers and passengers.

Solution

The data team tested two modeling architectures and compared their performance. Both models produced acceptable results, with the Random Forest model showing slightly better predictive accuracy. Based on these results, the team recommends beta testing the model with taxi drivers to gather additional feedback.

Details

Behind the data

- The data team hypothesized that a trip’s route, predicted fare, and time of day could have a strong influence on tipping behavior.
- After building and testing the models, the results confirmed that these factors are indeed predictive, with the final model achieving an F1 score of 0.7235.

Results Summary

The final algorithm can effectively predict which riders are likely to be generous tippers, demonstrating solid precision, recall, F1 score, and overall accuracy. See the “Next Steps” section for recommendations on further improvements and actions.

	model	precision	recall	F1	accuracy
0	RF CV	0.674919	0.757312	0.713601	0.680233
0	RF test	0.675297	0.779091	0.723490	0.686538
0	XGB CV	0.673074	0.724487	0.697756	0.669669
0	XGB test	0.675660	0.747978	0.709982	0.678349

Image Alt-Text: F1 scores for random forest and XGboost models

Future model suggestions

- Gather more detailed driver and rider data, including historical tipping behavior. Consider using K-means clustering to identify patterns and derive additional insights from the data.

Next Steps

As a next step, the Automatidata data team can present the model results to the New York City Taxi and Limousine Commission and suggest that the model be used as an indicator of tipping behavior. Additional data, however, would be required to achieve substantial improvements in predictive performance.